

MATH 472: INTRODUCTION TO TOPOLOGY

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Fall 2026

Last updated September 21, 2026.

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What is topology?

At its core, topology is the study of continuity, not only continuity of functions, but also continuous deformations of spaces, shapes, and surfaces. The popular conception of topology (Möbius strips, coffee cups = donuts, knots, etc.), however, fails to see what Kyle Ormsby calls the "...stunning breadth and extreme utility of topology in contemporary mathematics." [Orm, Week 1 Lecture Notes]. Here is a woefully incomplete list of places in mathematics and other fields of science where topology arises:

- (1) Manifold geometry: the study of manifolds and higher-dimensional surfaces requires passage from local regions to global geometry. Questions about gluing and connection.
- (2) Algebraic geometry: the Zariski topology, intersection theory, moduli spaces, complex varieties as topological spaces, etc.
- (3) Number Theory: there's a whole set of number-theoretic metrics out there called p-adic metrics, whereby two numbers are close to each other if they share the same divisors. Number theorists do analysis using this measure of distance.
- (4) Dynamical Systems and Flows: long-term behavior of differential equations, attractors, and qualitative features that persist under perturbation.
- (5) Combinatorics of Geometric Objects: graphs, posets, simplicial complexes; topological invariants constrain which objects exist and how they behave.
- (6) Topological Data Analysis: studying high-dimensional point clouds via their topology and using this to create summary statistics about data sets with homology.
- (7) Physics: Phase spaces, gauge symmetry, and homotopy theory in high energy physics.
- (8) Paleontology (and biology): I just read a book about modern paleontology and lots of the ideas therein seemed decidedly topological (although I doubt the author thought about them like that!). For example, distinguishing species of dinosaur by the number of holes in their skulls, or grouping species based on the number of shared characteristics (this is a notion of distance – hence continuity!)

The breadth of applications here implies that topology is a very abstract subject, and that is true in many ways. That's how topology shows up, but here's some of the ideas that characterize modern topology.

- (1) **Properties over construction:** when we study objects that are the same under continuous deformation, it often doesn't matter exactly how they are built. What we want to know is that no matter how you build them, they have the same topological properties. This leads naturally to the idea of category theory and universal properties, where we will *define* objects by the properties they have rather than by their construction.
- (2) **Topological equivalence:** there are many different ways to define a circle. Is it the image of $t \mapsto e^{2\pi i t}$ in $\mathbb{C}$? Or is it the image of $\theta \mapsto (\cos \theta, \sin \theta)$ in $\mathbb{R}^2$? Or is it what results from taking the unit interval

$[0, 1]$ and gluing 0 to 1? These are all really the same thing, and we say that they are topologically equivalent. The object that we really want to study is independent of the model chosen. There are multiple ways to measure topological equivalence, some more rigid than others.

- (3) **Invariants:** Given many different topologically equivalent avatars for the same object, we need some way to make differentiate between objects even when they're presented differently. Enter invariants: an invariant is a piece of data that stays the same (is "invariant") when replacing an object by a topologically equivalent one. We can use invariants to tell objects apart: if object X has invariant $I(X) = 1$ and object Y has invariant $I(Y) = 0$, then they cannot be topologically equivalent.

Chapter 1

Topological Spaces and Continuous Functions

1.1 Metric Spaces

The first principle above is properties over construction. With that in mind, our first task is to extract the properties of continuity that we learned from analysis and determine which properties determine them. We did this a little bit with the property of distance on the worksheet.

Definition 1.1.1. Let X be a set. A *metric* on X is a function $d: X \times X \rightarrow [0, \infty)$ with the following properties:

- For any $x \in X$, $d(x, x) = 0$.
- (positivity) For any $x, y \in X$, if $x \neq y$, then $d(x, y) > 0$
- (symmetry) For any $x, y \in X$, $d(x, y) = d(y, x)$
- (triangle inequality) For any $x, y, z \in X$, $d(x, z) \leq d(x, y) + d(y, z)$

Definition 1.1.2. A *metric space* is a pair (X, d) of a set X and a metric d on X .

Example 1.1.3. Here are some examples of metric spaces.

- (a) The metric space $(\mathbb{R}^n, d)$ where d is the standard Euclidean metric. For vectors $\vec{u}$ and $\vec{v}$, the distance between them is the length of their difference:

$$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|.$$

In $\mathbb{R}^2$ with vectors $\vec{u} = (u_0, u_1)$ and $\vec{v} = (v_0, v_1)$, we get

$$d(\vec{u}, \vec{v}) = \sqrt{(u_0 - v_0)^2 + (u_1 - v_1)^2}.$$

- (b) The taxicab metric on $\mathbb{R}^n$.
- (c) The L^p metric on $\mathbb{R}^n$.
- (d) The L^∞ metric on $\mathbb{R}^n$.
- (e) The set of binary strings of length n and Hamming distance: the number of positions at which the strings differ.

(f) The discrete metric on any space.

Given a notion of distance, we can talk about open balls in a metric space.

Definition 1.1.4. Let (X, d) be a metric space. Let $x \in X$ and let $r > 0$ be a positive real number. The *open ball of radius r around x* is the set

$$B_r(x) := \{y \in X \mid d(x, y) < r\}.$$

We saw some examples of this on the worksheet.

Example 1.1.5. In the hamming distance example, the open ball of radius 2 around any string is the set of all strings that differ from it by at most one character. The open ball of radius 1 around a given string is just the string itself (remember that it's a strict $<$).

Definition 1.1.6. Let (X, d) be a metric space. An *open set* in X is a set U such that for all $x \in U$, there exists some $r > 0$ such that $B_r(x) \subseteq U$.

Lemma 1.1.7. The open ball $B_r(x)$ is an open set.

Proof. Let $y \in B_r(x)$. Then $d(x, y) < r$ by the definition of an open ball. Let $\varepsilon = r - d(x, y)$. We claim that $B_\varepsilon(y) \subseteq B_r(x)$. Let $z \in B_\varepsilon(y)$. Then

$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \\ &< d(x, y) + (r - d(x, y)) \\ &= r. \end{aligned}$$

Therefore, $d(x, z) < r$, so $z \in B_r(x)$. Hence, $B_\varepsilon(y) \subseteq B_r(x)$. □

Example 1.1.8. Draw some pictures of open sets in $\mathbb{R}^2$ with various metrics.

Question 1.1.9. Are there sets in $\mathbb{R}^2$ that are open in the Euclidean metric that are not open in the taxicab metric, or vice versa?

Example 1.1.10. Every subset of $\mathbb{R}^2$ is open for the discrete metric, because the open balls of radius 1 are just points!

1.2 Bases for a Topology

A topology on a set X is the collection of open sets on X (these must follow some rules, which we'll get to soon). For metric spaces, we can define the open sets by those sets U such that around each point $x \in U$, there is an open ball $B_\varepsilon(x)$ completely contained in U . This suggests that the important thing in the topology is not just the open sets, but the open balls in particular. The notion of a basis generalizes the collection of open balls, without any reference to distances.

Recall that if X is a set, then $\mathcal{P}(X)$ is the powerset of X : the set of all subsets of X .

Definition 1.2.1. Let X be a set. A (topological) basis for X is a collection $\beta \subseteq \mathcal{P}(X)$ of subsets of X satisfying the following properties:

- *Covering.* For each $x \in X$, there exists some $B \in \beta$ such that $x \in B$.
- *Filtered.* For all $B_1, B_2 \in \beta$ and all $x \in B_1 \cap B_2$, there exists $B_3 \in \beta$ such that $x \in B_3$ and $B_3 \subseteq B_1 \cap B_2$.

Another way to say the covering property is that the union of all sets in β is X . In symbols:

$$\bigcup_{B \in \beta} B = X.$$

Example 1.2.2. The set of open balls in a metric space forms a basis for that topology. You will prove this on your homework. The proof uses similar ideas to the ones that we used in 1.1.7.

Example 1.2.3. For $a, b \in \mathbb{Z}$ with $a \neq 0$, let $S(a, b) = \{an + b \mid n \in \mathbb{Z}\}$. This is called an *arithmetic progression*. Then a topological basis on $\mathbb{Z}$ is given by the set of all arithmetic progressions

$$\beta = \{S(a, b) \mid a, b \in \mathbb{Z}\}.$$

Let's check that this is a basis:

- (covering) Let $n \in \mathbb{Z}$. We must find some arithmetic progression $S(a, b) \in \beta$ such that $n \in S(a, b)$. One option is $S(1, n)$, which contains n because $n = 0(1) + n$.
- (filtered) Suppose that we have basis sets $S(a, b)$ and $S(c, d)$. Let $x \in S(a, b) \cap S(c, d)$. We want to find a basic open set that contains x and is contained in both $S(a, b)$ and $S(c, d)$. The arithmetic progression $S(\text{lcm}(a, c), x)$ is such a basic open set, and $S(\text{lcm}(a, c), x) \subseteq S(a, b) \cap S(c, d)$.

The topology generated by this basis (see below) is called the *Furstenberg topology*. You can use this topology to give a proof of the infinitude of primes different from Euler's proof.

Given a basis, we can define open sets just as we did with the open balls, but with no reference whatsoever to a metric.

Definition 1.2.4. Let X be a set with a basis β . An subset $U \subseteq X$ is *open in X* if for each $x \in U$, there exists some $B \in \beta$ such that $x \in B$ and $B \subseteq U$.

Example 1.2.5. Let X be a set with a topological basis β . Then $\emptyset$ is open (vacuously) and X is open (by the covering axiom).

With the notion of a topological basis, we have axiomatized the idea of open balls to something more general than metric spaces. But we can go even further, and axiomatize the open sets themselves. Let's first think about some of the properties of open sets.

Lemma 1.2.6. Let X be a set with topological basis β . Let I be a set, and for each $i \in I$, let $U_i \subseteq X$ be an open set in X . Then the union U of all of the U_i ,

$$U := \bigcup_{i \in I} U_i$$

is open as well.

Proof. Let $U = \bigcup_{i \in I} U_i$. We must show that for each $x \in U$, there is some $B \in \beta$ such that $x \in B$ and $B \subseteq U$. To that end, note that if $x \in U$, then $x \in U_j$ for some j . Since U_j is open, there is some $B \in \beta$ such that $x \in B$ and $B \subseteq U_j$. But note that $U_j \subseteq U$ (U_j is a subset of the union of all of the U_i s), and therefore this B is the one we search for. In particular, $x \in B$ and $B \subseteq U_j \subseteq U$. $\square$

Lemma 1.2.7. Let X be a set with topological basis β . Let U and V be open sets in X . Then $U \cap V$ is open.

Proof. To show that $U \cap V$ is open, we must pick an arbitrary $x \in U \cap V$ and then find some $B \in \beta$ such that $x \in B$ and $B \subseteq U \cap V$.

To that end, let $x \in U \cap V$. Then $x \in U$, and U is open, so there exists some $B_1 \in \beta$ such that $x \in B_1$ and $B_1 \subseteq U$.

Similarly, $x \in V$, and V is open, so there exists some $B_2 \in \beta$ such that $x \in B_2$ and $B_2 \subseteq V$.

Now we have $x \in B_1$ and $x \in B_2$, which means that $x \in B_1 \cap B_2$. By the filtering axiom of a basis, this means that there exists some $B_3 \in \beta$ such that $x \in B_3$ and $B_3 \subseteq B_1 \cap B_2$.

This is the B that we were searching for: since $B_1 \subseteq U$ and $B_2 \subseteq V$, we have

$$B_1 \cap B_2 \subseteq U \cap V.$$

Then $B_3 \subseteq B_1 \cap B_2$ implies that $B_3 \subseteq U \cap V$. We also have $x \in B_3$, so we're done! $\square$

Lemma 1.2.8. Let X be a set with topological basis β . Let $U_1, \dots, U_n$ be open sets in X . Then the intersection

$$\bigcap_{i=1}^n U_i$$

is open.

Proof. Proof by induction on n . For $n = 0$ the empty intersection is all of X , which we already showed was open in 1.2.5. For $n = 1$, the intersection of one open set is open again. For $n = 2$, this is the previous lemma 1.2.7. This handles the base cases.

For the inductive step, assume that the intersection of any $n - 1$ open sets is open. Let $V = \bigcap_{i=1}^{n-1} U_i$. Then by the previous lemma 1.2.7, $U_n \cap V$ is the intersection of two opens, and is therefore open. But

$$U_n \cap V = U_n \cap \bigcap_{i=1}^{n-1} U_i = \bigcap_{i=1}^n U_i,$$

which is what we were trying to prove is open. $\square$

Example 1.2.9. We can't get away with arbitrary intersections of open sets being open. For example, consider the intersection of all open balls of radius $1/n$ in the Euclidean topology on $\mathbb{R}^2$:

$$\bigcap_{n=1}^{\infty} B_{\frac{1}{n}}((0,0)).$$

This set contains just one point: the origin. And a single point isn't open.

1.3 Topologies

Knowing what we know about the open sets in X with respect to a basis β , we can axiomatize the open sets.

Definition 1.3.1. Let X be a set. A *topology* τ on X is a collection $\tau \subseteq \mathcal{P}(X)$ of subsets of X such that

- Arbitrary unions of sets in τ are again in τ . In symbols: if I is a set and $U_i \in \tau$ for all $i \in I$, then

$$\bigcup_{i \in I} U_i \in \tau.$$

- Finite intersections of sets in τ are again in τ . In symbols: if $U_1, \dots, U_n \in \tau$ for some nonnegative integer n , then

$$\bigcap_{i=1}^n U_i \in \tau.$$

We call the elements of τ the *open sets* of the topology.

Definition 1.3.2. A *topological space* is a pair (X, τ) of a set X and a topology τ on X .

Remark 1.3.3. Some authors also demand the axiom that $\emptyset \in \tau$ and $X \in \tau$. But these follow from the axioms given above! Can you see why?*

This is even more abstract than the abstract bases we talked about earlier! But we already have some examples given by sets with bases.

*The answer is that the union of zero sets is the empty set, so $\emptyset \in \tau$ follows from the unions axiom. Similarly, the intersection of zero sets is all of X , so $X \in \tau$ follows from the finite intersection axiom.

Theorem 1.3.4. Let X be a set with a topological basis β . Let τ be the set

$$\tau := \left\{ U \subseteq X \mid U \text{ is open with respect to the basis } \beta \right\}.$$

We call this the topology generated by the basis β . Then τ is a topology on X .

Proof. This is what we proved in the last section. See 1.2.6 and 1.2.8 □

But there are many, many more examples.

Example 1.3.5. On any set X , the *discrete topology* is the topology τ_{disc} where every set is open. Then τ_{disc} is the whole powerset of X : $\tau_{\text{disc}} := \mathcal{P}(X)$

Example 1.3.6. On any set X , the *indiscrete topology* or the *concrete topology* is the topology $\tau_{\text{conc}} := \{\emptyset, X\}$. That is, the only open sets in this topology are empty or the whole set.

Example 1.3.7. The set $\{0, 1\}$ has a topology given by

$$\tau = \{\emptyset, \{1\}, \{0, 1\}\}.$$

This is not the discrete topology (because $\{0\}$ isn't open), and it's not the concrete topology (because $\{1\}$ is open). So this is a genuinely new topology on $\{0, 1\}$. Are there any others?

Example 1.3.8. Given a topological space (X, τ) and a subset $A \subseteq X$, we can build a new topology on A by saying that the open sets in A are the sets of the form $U \cap A$ for some open set U of X . In symbols, the topology on A is

$$\tau' = \left\{ U \cap A \mid U \in \tau \right\}.$$

This is called the *subspace topology*.

Example 1.3.9. Cofinite topology on $\mathbb{N}$. The topology τ_{cof} is

$$\tau_{\text{cof}} := \left\{ U \subseteq \mathbb{N} \mid \mathbb{N} \setminus U \text{ is finite} \right\} \cup \{\emptyset\}.$$

In words, the nonempty open sets in this topology are the subsets of $\mathbb{N}$ whose complement is finite. Note that this is different than infinite sets being open! The odd numbers is an infinite set whose complement (the even numbers) is still infinite.

Example 1.3.10. The poset topology. Let $(P, \leq)$ be a poset. A subset $U \subseteq P$ is called *upwards closed* if for all $x \in U$ and $y \in P$ such that $x \leq y$, then $y \in U$ as well. In other words, U is upwards closed if $x \in U$ implies that $y \in U$ for all $y \geq x$.

Let τ be the set of all upwards closed subsets of P . Then τ is a topology on P .

$$\tau = \left\{ U \subseteq P \mid U \text{ is upwards closed} \right\}.$$

Frequently in the previous few sections, we made use of open balls around a point. But when we're talking about general topological spaces, we no longer have open balls. So we use the idea of a neighborhood instead. This is very important vocabulary that we will use often!

Definition 1.3.11. Let (X, τ) be a topological space and let $x \in X$. An (*open*) *neighborhood* of x is an open set $U \in \tau$ such that $x \in U$.

1.4 Closed Sets

Another important idea in topology is the notion of a closed set.

Definition 1.4.1. Let (X, τ) be a topological space. A subset $K \subseteq X$ is *closed* if $X \setminus K$ is open.

Example 1.4.2. Since $\emptyset$ and X are always open, then $X = X \setminus \emptyset$ and $\emptyset = X \setminus X$ are always closed as well. This illustrates that closed and open are not mutually exclusive!

It is false to say that if a set is not open, then it is closed, as the following example shows.

Example 1.4.3. In the Euclidean topology on $\mathbb{R}$, open intervals (a, b) are open sets, and closed intervals $[a, b]$ are closed sets. But there are intervals of the form $[a, b)$ and $(a, b]$ that are neither open nor closed.

Example 1.4.4. In the cofinite topology on $\mathbb{N}$, the closed sets are either finite sets or all of $\mathbb{N}$.

You can also define a topology by specifying the closed sets. In fact, this is probably the better way to define some topologies, like the cofinite topology.

Theorem 1.4.5. Let X be a set. Let $\kappa \subseteq \mathcal{P}(X)$ be a collection of subsets of X such that

- (a) κ is closed under finite unions: if $K_1, \dots, K_n \in \kappa$, then $\bigcup_{i=1}^n K_i \in \kappa$.
- (b) κ is closed under arbitrary intersections: if I is a set and $K_i \in \kappa$ for each $i \in I$, then $\bigcap_{i \in I} K_i \in \kappa$.

Then

$$\tau = \{X \setminus K \mid K \in \kappa\}.$$

is a topology on X .

Proof. To prove that τ is a topology on X , we must show that τ is closed under finite intersections and arbitrary unions.

- (τ is closed under finite intersection) Let $U_1, \dots, U_n \in \tau$. Because every set in τ is the complement of a set in κ , then for each $i = 1, \dots, n$ there is some $K_i \in \kappa$ such that $U_i = X \setminus K_i$. Then

$$\bigcap_{i=1}^n U_i = \bigcap_{i=1}^n (X \setminus K_i) = X \setminus \left(\bigcup_{i=1}^n K_i \right).$$

Since κ is closed under finite union, this shows that the intersection of all the U_i s is again a complement of a set in κ . Hence, $\bigcap_{i=1}^n U_i \in \tau$.

- (τ is closed under arbitrary union) Let I be a set and let $U_i \in \tau$ for each $i \in I$. Because every set in τ is the complement of a set in κ , then for each $i \in I$, there is some $K_i \in \kappa$ such that $U_i = X \setminus K_i$. Then

$$\bigcup_{i \in I} U_i = \bigcup_{i \in I} (X \setminus K_i) = X \setminus \left(\bigcap_{i \in I} K_i \right).$$

Since κ is closed under arbitrary intersection, this shows that the union of all of the U_i s is again a complement of a set in κ . Hence, $\bigcup_{i \in I} U_i \in \tau$.

This shows that τ is both closed under finite intersection and arbitrary union, and hence τ is a topology. $\square$

1.4.1 Infinitely Many Prime Numbers

We can return to Example 1.2.3 and use it to prove that there are infinitely many prime numbers. Recall that we can define a topology on $\mathbb{Z}$ by stating that the basic open sets are the arithmetic progressions $S(a, b)$ with $a \neq 0$, where

$$S(a, b) = \{an + b \mid n \in \mathbb{Z}\}$$

That is, a set $U \subseteq \mathbb{Z}$ is open in this topology if for all $x \in U$, there is some $S(a, b)$ such that $x \in S(a, b)$ and $S(a, b) \subseteq U$.

Lemma 1.4.6. $S(a, b)$ is a closed set.

Proof. We must write $S(a, b)$ as the complement of an open set to show that it is closed.

$$S(a, b) = \mathbb{Z} \setminus \left(\bigcup_{i=1}^{a-1} S(a, b+i) \right)$$

□

This is an example of a topology in which there are many sets that are both open and closed!

Lemma 1.4.7. Every nonempty open set in this topology is infinite.

Proof. If U is open and nonempty, then there is some $x \in U$. Since U is open, then there is some $S(a, b)$ such that $x \in S(a, b)$ and $S(a, b) \subseteq U$. Since $S(a, b)$ is an infinite set, then U must be infinite as well. □

Theorem 1.4.8. There are infinitely many prime numbers.

Proof. Proof by contradiction. Assume that there are finitely many prime numbers.

By Lemma 1.4.7, the set $\{-1, +1\}$ cannot be open. Therefore, $\mathbb{Z} \setminus \{-1, +1\}$ cannot be a closed set. However, we can write

$$\mathbb{Z} \setminus \{-1, +1\} = \bigcup_{p \text{ prime}} S(p, 0),$$

which by Lemma 1.4.6 is a finite union of closed sets, and therefore closed. (Remember we are operating under the assumption of finitely many primes.) This contradicts the fact that $\mathbb{Z} \setminus \{-1, +1\}$ cannot be closed, so our assumption must be false and there are infinitely many primes. □

1.5 Continuous Functions

Topology appears whenever we have a notion of closeness, whether we define it via a metric or a topological basis or a topology.

Definition 1.5.1. Let X and Y be sets, and let $f: X \rightarrow Y$ be a function.

(a) Let $B \subseteq Y$. The *preimage* of B under f is the set

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}.$$

(b) Let $A \subseteq X$. The *image* of A under f is the set

$$f(A) = \{f(a) \mid a \in A\}.$$

Definition 1.5.2. Let X and Y be topological spaces, and let $f: X \rightarrow Y$ be a function. The function f is *continuous* if for each open subset V of Y , $f^{-1}(V)$ is open.

Let's see how this definition of continuity matches up with the definition of continuity you learned in analysis.

Proposition 1.5.3. *Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function, meaning that for all $x_0 \in \mathbb{R}$ and all $\varepsilon > 0$, there is some $\delta > 0$ such that if $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.*

Then f is continuous in the sense of topology (1.5.2).

Proof. Assume that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function. Let $V \subseteq \mathbb{R}$ be an open set. We want to show that $f^{-1}(V)$ is open.

To show that $f^{-1}(V)$ is open, we must pick some point $x_0 \in f^{-1}(V)$ and show that we can fit an open ball around x_0 entirely inside $f^{-1}(V)$.

Since V is open, we can fit an open ball around any point inside of V . In particular, $f(x_0) \in V$ by the definition of $x_0 \in f^{-1}(V)$. Therefore, there is an open ball $B_\varepsilon(f(x_0)) \subseteq V$.

The analysis definition of continuity shows us that if $x \in B_\delta(x_0)$, then $f(x) \in B_\varepsilon(f(x_0))$. In other words,

$$B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0))).$$

Since $B_\varepsilon(f(x_0)) \subseteq V$, we see then that $f^{-1}(B_\varepsilon(f(x_0))) \subseteq f^{-1}(V)$. Therefore,

$$B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0))) \subseteq f^{-1}(V).$$

This finds an open set around x_0 that is contained in $f^{-1}(V)$. Since our choice of x_0 was arbitrary, $f^{-1}(V)$ is open. Therefore, f is continuous in the sense of topology. $\square$

Exercise 1.5.4. Prove that if $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous in the sense of topology (1.5.2), then $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous in the sense of analysis.

Example 1.5.5. • identity functions are continuous

- every function whose domain is discrete is continuous
- from X one topology to X with another topology
- taxicab geometry and regular geometry

1.5.1 Building Continuous Functions

How can you go about building continuous functions from one space to another? There are a couple of standard techniques.

Proposition 1.5.6. *Let X and Y be topological spaces and let $y \in Y$. Then the constant function $f: X \rightarrow Y$ defined by $f(x) = y$ for all $x \in X$ is continuous.*

Proof. Let $V \subseteq Y$ be open. If $y \in V$, then $f^{-1}(V) = X$, which is open. If $y \notin V$, then $f^{-1}(V) = \emptyset$, which is also open. Therefore, the preimage of any open in Y is open, which means that f is continuous. $\square$

Proposition 1.5.7. *Let X, Y , and Z be topological spaces with continuous functions $f: X \rightarrow Y$ and $g: Y \rightarrow Z$. Then $g \circ f: X \rightarrow Z$ is continuous.*

Proof. To show that $g \circ f: X \rightarrow Z$ is continuous, we need to show that for each open set $W \subseteq Z$, $(g \circ f)^{-1}(W)$ is open in X . But notice that

$$(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W)).$$

Since g is continuous and W is open, then $g^{-1}(W)$ is open. Since f is continuous and $g^{-1}(W)$ is open, then $f^{-1}(g^{-1}(W))$ is open. Therefore, $(g \circ f)^{-1}(W)$ is open, so $g \circ f$ is continuous. $\square$

Lemma 1.5.8. Let X be a topological space with a subspace A . Let $i: A \hookrightarrow X$ be the inclusion function. Then $i: A \rightarrow X$ is continuous.

Proof. Let $U \subseteq X$ be open. Then $i^{-1}(U) = U \cap A$, which is open in A by the definition of the subspace topology. $\square$

Proposition 1.5.9. Let $f: X \rightarrow Y$ be a continuous function between topological spaces. Let $A \subseteq X$ be a subspace with the subspace topology. Then if we restrict the domain of f to A , it is still continuous.

Proof. Let $i: A \hookrightarrow X$ be the inclusion function. Then restricting the domain of f to A is the same as composing $f \circ i$. Since f is continuous (by assumption) and i is continuous (by previous lemma), then $f \circ i$ is continuous by a previous proposition. $\square$

Lemma 1.5.10. Let X be a topological space with subspace A . Give A the subspace topology.

- (a) If A is open in X and $U \subseteq A$ is open in A , then U is open in X .
- (b) If A is closed in X and $K \subseteq A$ is closed in A , then K is closed in X .

Theorem 1.5.11 (Pasting). Let X be a topological space such that $X = A \cup B$, where A and B are closed in X . Then let $f: A \rightarrow Y$ and $g: B \rightarrow Y$ be continuous functions, where A and B are given the subspace topology. If $f(x) = g(x)$ for all $x \in A \cap B$, then f and g can be combined to give a continuous function $h: X \rightarrow Y$ defined by

$$h(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B. \end{cases}$$

Proof. We will use the formulation of continuity where $f: Y \rightarrow Z$ is continuous if and only if for each closed subset $K \subseteq Z$, $f^{-1}(K)$ is closed in Y .

Let C be a closed set in Y . Then $h^{-1}(C) = f^{-1}(C) \cup g^{-1}(C)$. Since f is continuous, $f^{-1}(C)$ is closed in A , and therefore closed in X by the previous lemma. Since g is continuous, $g^{-1}(C)$ is closed in B , and therefore closed in X by the previous lemma. Hence, $h^{-1}(C)$ is the union of two closed sets, and therefore closed. So h sends closed sets to closed sets and is therefore continuous. $\square$

Example 1.5.12. The function

$$h(x) = \begin{cases} 2x & \text{if } x \leq 0 \\ x/2 & \text{if } x \geq 0 \end{cases}$$

is defined by two functions $f: (-\infty, 0] \rightarrow \mathbb{R}$, $f(x) = 2x$ and $g: [0, \infty) \rightarrow \mathbb{R}$, $g(x) = x/2$ on two closed sets $A = (-\infty, 0]$ and $B = [0, \infty)$. The functions f and g agree on $A \cap B = \{0\}$. The pasting theorem guarantees that h is continuous.

1.5.2 Homeomorphisms

Definition 1.5.13. A function $f: X \rightarrow Y$ between topological spaces is a *homeomorphism* if

- f is continuous
- f is a bijection with inverse function g
- the inverse function $g: Y \rightarrow X$ is continuous

Definition 1.5.14. If $f: X \rightarrow Y$ is a homeomorphism, then we say that X and Y are *homeomorphic* and write $X \cong Y$.

Example 1.5.15. The open interval $(0, 1)$ is homeomorphic to any other open interval (a, b) via

$$f(x) = xb + (1 - x)a = a + x(b - a).$$

The inverse is

$$g(x) = \frac{x - a}{b - a}.$$

Example 1.5.16. The closed interval $[0, 1]$ is homeomorphic to $[a, b]$ via the same formulas above.

Example 1.5.17. The open interval $(0, 1)$ is homeomorphic to $\mathbb{R}$ via

$$f(x) = \frac{1}{x(x - 1)}.$$

Example 1.5.18. Stereographic projection. The unit sphere S^2 is the set of points a distance of 1 from the origin in $\mathbb{R}^3$.

$$S^2 = \left\{ (x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1 \right\}.$$

Let $N = (0, 0, 1)$ be the north pole of S^2 . Then $S^2 \setminus \{N\}$ is homeomorphic to $\mathbb{R}^2$ via the functions $f: S^2 \setminus \{N\} \rightarrow \mathbb{R}^2$

$$f(x, y, z) = \left(\frac{x}{1 - z}, \frac{y}{1 - z} \right),$$

and $g: \mathbb{R}^2 \rightarrow S^2 \setminus \{N\}$

$$g(x, y) = \left(\frac{2x}{1 + x^2 + y^2}, \frac{2y}{1 + x^2 + y^2}, \frac{-1 + x^2 + y^2}{1 + x^2 + y^2} \right).$$

You can check that these are inverse functions and are continuous. There's also a nice geometric description of this process: for a point $P = (x, y, z)$ on $S^2 \setminus \{N\}$, draw the straight line in $\mathbb{R}^3$ through N and P . This will intersect the plane $(z = 0)$ in one point, which is the point $f(P)$ in $\mathbb{R}^2$. Conversely, given a point in the plane $(z = 0)$, you can draw a line through it and N to find a unique point on the sphere.

It's very important that the inverse function is also continuous! It's not automatic that the inverse is continuous, as the following example shows.

Example 1.5.19. $[0, 1) \rightarrow S^1$ given by $t \mapsto e^{2\pi it}$ is continuous and a bijection, but no continuous inverse. In particular, $[0, \frac{1}{2})$ is open in the interval $[0, 1)$ under the subspace topology but its preimage in the circle S^1 is not open in the subspace topology on $\mathbb{C}$; it includes $z = 1$ along the real axis.

Definition 1.5.20. Let $f: X \rightarrow Y$ be a function between topological spaces. Then we say that f is

- *open* if for each open $U \subseteq X$, $f(U)$ is open in Y .
- *closed* if for each closed $K \subseteq X$, $f(K)$ is closed in Y .

There's another useful characterization of homeomorphisms.

Lemma 1.5.21. Let $f: X \rightarrow Y$ be a bijection with inverse $g: Y \rightarrow X$, and let $A \subseteq X$ be any subset. Then

$$g^{-1}(A) = f(A).$$

Proof.

$$g^{-1}(A) = \{y \in Y \mid g(y) \in A\} = \{f(g(y)) \in Y \mid g(y) \in A\} = \{f(x) \in Y \mid x \in A\} = f(A).$$

□

Proposition 1.5.22. Let $f: X \rightarrow Y$ be a continuous function between topological spaces. Then the following are equivalent:

(a) $f: X \rightarrow Y$ is a homeomorphism

(b) $f: X \rightarrow Y$ is an open bijection

Proof. ($\Leftarrow$). Assume that $f: X \rightarrow Y$ is an open bijection. Let $g: Y \rightarrow X$ be the inverse function of f . To show that f is a homeomorphism, we must show that $g: Y \rightarrow X$ is continuous.

To do so, let $U \subseteq X$ be open. We want to show that $g^{-1}(U)$ is open in Y . But notice that by the lemma, $g^{-1}(U) = f(U)$, which is open because f is an open function.

($\Rightarrow$). Assume that $f: X \rightarrow Y$ is a homeomorphism, and let $g: Y \rightarrow X$ be the inverse function to f . To show that f is an open function, we must show that for any $U \subseteq X$, $f(U)$ is open. But by the lemma, $f(U) = g^{-1}(U)$, which is open because g is continuous. $\square$

This shows that a homeomorphism $f: X \rightarrow Y$ not only sends points in X bijectively to points in Y , but also open sets in X bijectively to open sets in Y . For that reason, whenever there is any property of X that can be expressed entirely in terms of the topology of X (that is, entirely in terms of the open sets of X), then that property applies to Y as well. Such a property of X is called a *topological property*.

Definition 1.5.23. A *topological property* P is a property of topological spaces such that a space X has property P and $X \cong Y$ implies that Y has property P as well.

In other words, a topological property is a property that is *homeomorphism invariant*.

1.6 Comparing Topologies

This is a good time to bring up some vocabulary that comes up when comparing topologies on the same set.

Definition 1.6.1. Suppose that X is a set and that τ, τ' are two topologies on X .

(a) We say that τ' is *finer* than τ if $\tau' \supseteq \tau$.

(b) We say that τ' is *coarser* than τ if $\tau' \subseteq \tau$.

This vocabulary is meant to convey the analogy of gravel or dirt. If you think of the open sets of a topology as pebbles, and you smash them into bits, then the collection of open sets has been enlarged and the topology is said to have been made finer. So even though the term *finer* suggests some idea of “smaller-than,” the *larger* topology τ' is the finer one (because it has smaller open sets).

To be clear:

- If τ' is finer than τ , then every set that is open in τ is open in τ' .
- If τ' is coarser than τ , then every set that is open in τ' is open in τ .

Example 1.6.2. The discrete topology on a set X is finer than any other topology on X ; it is the finest topology.

Example 1.6.3. The indiscrete topology on a set X is coarser than any other topology on X ; it is the coarsest topology.

Example 1.6.4. The *lower limit topology* on $\mathbb{R}$ is the topology on $\mathbb{R}$ with a basis of open sets given by $[a, b)$. Let $\mathbb{R}_\ell$ denote the topological space that is the set $\mathbb{R}$ equipped with the lower limit topology, and let $\mathbb{R}_s$ denote $\mathbb{R}$ with the standard topology.

In $\mathbb{R}_\ell$, (a, b) is open because

$$(a, b) = \bigcup_{n=1}^{\infty} \left[a + \frac{1}{n}, b \right).$$

But $[a, b)$ is not open in $\mathbb{R}_s$. Therefore, $\mathbb{R}_\ell$ has more open sets, so it is finer than $\mathbb{R}_s$.

1.7 The Product Topology

How is the topology on $\mathbb{R}$ related to the topology on $\mathbb{R}^2$? One answer is that we can think of $\mathbb{R}$ as a subset of $\mathbb{R}^2$ (as the x -axis, say, or as any line), and that the subspace topology on this line gives the usual topology on $\mathbb{R}$. Another answer is that we can build the standard topology on $\mathbb{R}^2$ using the topology on $\mathbb{R}$ as a product topology.

Definition 1.7.1. Let X and Y be topological spaces. The *product topology* on $X \times Y$ is the topology with basis $U \times V$, where U is open in X and V is open in Y .

We also apply this definition to products of finitely many spaces.

Definition 1.7.2. Let $X_1, \dots, X_n$ be topological spaces. The *product topology* on $X_1 \times X_2 \times \dots \times X_n$ is the topology with basis given by sets of the form $U_1 \times U_2 \times \dots \times U_n$, where U_i is open in X_i for each $i = 1, 2, \dots, n$.

Example 1.7.3. Consider the product topology on $\mathbb{R} \times \mathbb{R}$, where $\mathbb{R}$ has the standard topology. A basic open set in the product topology is an open box of the form $(a, b) \times (c, d)$, and a set U is open in this topology if and only if for all $x \in U$, there is some open box B such that $x \in B$ and $B \subseteq U$.

This space is homeomorphic to $\mathbb{R}^2$ with the standard topology (wherein the basic open sets are open balls).

There are two important functions for any product $X \times Y$.

Definition 1.7.4. Let X and Y be sets (or spaces). The *projection functions* are the functions $p_1: X \times Y \rightarrow X$ and $p_2: X \times Y \rightarrow Y$ defined by

$$\begin{aligned} p_1(x, y) &= x \\ p_2(x, y) &= y. \end{aligned}$$

When X and Y are topological spaces, these functions have nice topological properties.

Proposition 1.7.5. Let X and Y be spaces and give $X \times Y$ the product topology. Then the projections p_1 and p_2 are continuous and open.

Proof. We will prove this for p_1 ; the case for p_2 is basically the same.

To show that $p_1: X \times Y \rightarrow X$ is continuous, we want to show that for any $U \subseteq X$ open, $p_1^{-1}(U)$ is open in $X \times Y$. But $p_1^{-1}(U) = U \times Y$, which is open in $X \times Y$ since U and Y are both open in X and Y , respectively. Hence, p_1 is continuous.

To show that $p_1: X \times Y \rightarrow X$ is open, we must show that for any open set $W \subseteq X \times Y$, $p_1(W)$ is open in X . It suffices to show this when W is a basic open set (exercise). If W is a basic open set, then $W = U \times V$ where U is open in X and V is open in Y . Then

$$p_1(W) = p_1(U \times V) = U,$$

which is open in X . □

The projection functions help us find continuous functions into the product.

Proposition 1.7.6. Let X, Y , and Z be topological spaces. A function $f: Z \rightarrow X \times Y$ is continuous if and only if both $p_1 \circ f$ and $p_2 \circ f$ are continuous.

Proof. ($\implies$). If f is continuous, then $p_1 \circ f$ and $p_2 \circ f$ are both continuous by 1.5.7.

($\impliedby$). To show that f is continuous, we must show that $f^{-1}(U \times V)$ is open in Z for each basic open set $U \times V$ of $X \times Y$ by a homework problem.

$$f^{-1}(U \times V) = (p_1 \circ f)^{-1}(U) \cap (p_2 \circ f)^{-1}(V),$$

and since $p_1 \circ f$ and $p_2 \circ f$ are both continuous, then the right hand side is open. Therefore, $f^{-1}(U \times V)$ is open in Z . $\square$

There is a neat characterization of the product topology in terms of the projections.

Proposition 1.7.7. *The product topology on $X \times Y$ is the coarsest topology on $X \times Y$ for which both p_1 and p_2 are continuous.*

Proof. Suppose that τ is any topology on $X \times Y$ topology such that both projections are continuous with respect to τ . Then for open sets $U \subseteq X$ and $V \subseteq Y$, $p_1^{-1}(U) \cap p_2^{-1}(V)$ must be open in $X \times Y$. But

$$p_1^{-1}(U) \cap p_2^{-1}(V) = U \times V.$$

Therefore, in *any* topology such that both projections are continuous, $U \times V$ must be open for any open $U \subseteq X$ and $V \subseteq Y$. So any such topology contains the basic open sets of the product topology, and so τ must be finer than the product topology. In other words, the product topology is coarser than τ . $\square$

We can combine the previous two propositions into the following.

Theorem 1.7.8 (Universal Property of the Product). *Let X and Y be spaces. The product $X \times Y$ is the topological space with two functions $p_1: X \times Y \rightarrow X$ and $p_2: X \times Y \rightarrow Y$ such that for any other topological space Z and continuous functions $f: Z \rightarrow X$ and $g: Z \rightarrow Y$, there is a unique continuous function $h: Z \rightarrow X \times Y$ such that $p_1 \circ h = f$ and $p_2 \circ h = g$.*

$$\begin{array}{ccccc} & & Z & & \\ & \swarrow f & \downarrow \exists! h & \searrow g & \\ X & \xleftarrow{p_1} & X \times Y & \xrightarrow{p_2} & Y \end{array}$$

Chapter 2

Topological Properties

2.1 Categories

Category theory was introduced by Samuel Eilenberg and Saunders Mac Lane in the mid 1900s to put their work on algebraic topology on solid ground. Although it was originally motivated by topological considerations, category theory has become an incredibly useful tool in many areas of mathematics.

Category theory builds on and formalizes a philosophical shift that occurred in mathematics in the 1900s, namely that we should think not only about mathematical objects themselves, but also about the relations between them. These relations take many forms, but often they take the form of functions between sets with some kind of special property. Once we have the language of category theory, we can prove theorems about the categories themselves and apply them to different areas of mathematics systematically. In fact, many seemingly different ideas in different branches of math turn out to come from the same categorical constructions. In other words, category theory becomes a meta-mathematical tool that we can use to translate results from one area of mathematics to another.

We begin with a motivating example.

Example 2.1.1. One kind of mathematical object that shows up everywhere is sets. Sets are related by functions. Any function $f: A \rightarrow B$ has a domain A (also called the source) and a codomain B (also called the target). You can also compose functions: given $f: A \rightarrow B$ and $g: B \rightarrow C$, there is a new function called $g \circ f: A \rightarrow C$ given by $a \mapsto g(f(a))$ for all $a \in A$. Composition of functions is associative: if we have a third function $h: C \rightarrow D$, we have

$$(h \circ g) \circ f = h \circ (g \circ f).$$

For every set A , there is an identity function $\text{id}_A: A \rightarrow A$ given by $\text{id}_A(a) = a$. The special thing about the identity function is that for any function $f: A \rightarrow B$,

$$f \circ \text{id}_A = f,$$

and for any function $k: E \rightarrow A$,

$$\text{id}_A \circ k = k.$$

It is this kind of thing that the definition of a category seeks to generalize.

Definition 2.1.2. A *category* $\mathcal{C}$ is the following data:

- A collection^{*} of *objects* $\text{Ob}(\mathcal{C})$.

^{*}We say a *collection* instead of *set* because we will consider things like the category of all sets. But there is no set of all sets because that creates paradoxes (does the set of all sets contain itself?). So we say that we're allowed to have a collection of all sets, and that a collection is not a set, to sidestep this issue. For the most part, you never need to worry about this and you can just leave it to the logicians to sort it out.

- A collection of *morphisms* $\text{Mor}(\mathcal{C})$ such that each morphism $f \in \text{Mor}(\mathcal{C})$ has both a domain object and a codomain object. If $X, Y \in \text{Ob}(\mathcal{C})$, we write $\mathcal{C}(X, Y)$ [†] for the collection of morphisms with domain X and codomain Y . If $f \in \mathcal{C}(X, Y)$, we will also write $f: X \rightarrow Y$.
- For each $X \in \text{Ob}(\mathcal{C})$, there is an *identity morphism* $\text{id}_X: X \rightarrow X$.
- For any three objects $X, Y, Z \in \text{Ob}(\mathcal{C})$, there is a composition operation

$$\circ: \mathcal{C}(Y, Z) \times \mathcal{C}(X, Y) \rightarrow \mathcal{C}(X, Z),$$

that composes morphisms $g: Y \rightarrow Z$ and $f: X \rightarrow Y$ to form a new morphism $g \circ f: X \rightarrow Z$.

These data must satisfy the following properties:

- *Associativity*. Composition is associative. In more detail, given $f: X \rightarrow Y$, $g: Y \rightarrow Z$, and $h: Z \rightarrow W$,

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

- *Identity*. For any $f: X \rightarrow Y$, we have

$$f \circ \text{id}_X = f \quad \text{and} \quad \text{id}_Y \circ f = f.$$

While it looks like a lot to write down, the thing to remember about categories is this: there's objects and morphisms and composition of morphisms, and the composition works exactly like you'd expect if it were composition of functions. You've been working with categories for your whole career as mathematicians, without realizing it.

Example 2.1.3. We've already mentioned that there is a category Set of sets and functions. Similarly, there is also a category Fin , where the objects are finite sets and the morphisms are functions.

Example 2.1.4. The category $\text{Vect}_{\mathbb{R}}$ has objects $\mathbb{R}$ -vector spaces and morphisms linear transformations. Since linear transformations compose to give new linear transformations, and the identity functions $\text{id}_V: V \rightarrow V$ are linear transformations, this is well-defined as a category. There is also a category $\text{Vect}_{\mathbb{C}}$ of $\mathbb{C}$ -vector spaces.

We've been working in one particular category this entire class.

Example 2.1.5. The objects of the category Top are topological spaces. The morphisms in this category are continuous functions. This category is well-defined because composing two continuous functions gives another continuous function, and identity functions are continuous.

There are also examples that are less familiar.

Example 2.1.6. The empty category $\emptyset$ has no objects (i.e. $\text{Ob}(\emptyset) = \emptyset$) and no morphisms ($\text{Mor}(\emptyset) = \emptyset$). Composition is the empty function $\emptyset \rightarrow \emptyset$ and all properties are satisfied vacuously.

Example 2.1.7. The trivial category $\mathbf{1}$ has only one object $\text{Ob}(\mathbf{1}) = \{*\}$, and a single morphism $\text{id}_*: * \rightarrow *$. The only composition to define is $\text{id} \circ \text{id} = \text{id}$, and this gives the associative property and identity property too.

[†]You may have seen $\text{Hom}_{\mathcal{C}}(X, Y)$ in other classes for the same thing.

2.1.1 Commutative Diagrams

It is useful to draw morphisms in a category $\mathcal{C}$ diagrammatically. We may write $f: X \rightarrow Y$ to mean $f \in \mathcal{C}(X, Y)$, but we may also draw

$$X \xrightarrow{f} Y$$

to mean the same information. Now suppose we have another morphism

$$Y \xrightarrow{g} Z.$$

Then we can compose these to get a morphism

$$X \xrightarrow{g \circ f} Z.$$

We can put all of this information into a single *commutative diagram*

$$\begin{array}{ccccc} X & \xrightarrow{f} & Y & \xrightarrow{g} & Z \\ & \searrow & & \nearrow & \\ & & g \circ f & & \end{array}$$

The exact geometry of the diagram doesn't matter. It could equally well be drawn as a triangle

$$\begin{array}{ccc} & Y & \\ f \nearrow & & \searrow g \\ X & \xrightarrow{g \circ f} & Z \end{array}$$

and it would convey the same information.

We can use commutative diagrams to communicate the axioms of a category. For example, associativity is the diagram

$$\begin{array}{ccccc} & Y & \xrightarrow{h \circ g} & W & \\ f \nearrow & & \searrow g & \nearrow h & \\ X & \xrightarrow{g \circ f} & Z & & \end{array}$$

which says that we can paste together commutative diagrams to get commutative quadrilaterals. The identity axiom becomes

$$\begin{array}{ccc} X & \xrightarrow{\text{id}_X} & X \\ & \searrow f & \downarrow f \\ & & X \end{array} \quad \text{and} \quad \begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow f & \downarrow \text{id}_Y \\ & & Y. \end{array}$$

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